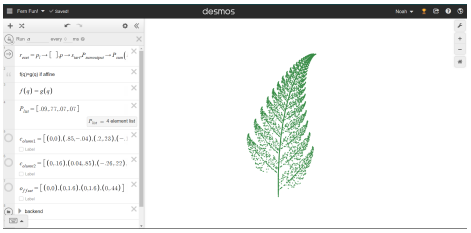

Figure 1: This graph creates a fractal called the "Barnsley Fern" using a chaos game on an iterated function system. Link: `https://www.desmos.com/calculator/iusw2jis9d`

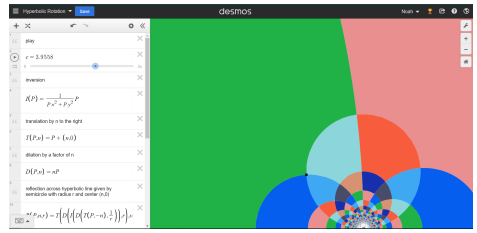

Figure 2: This graph is a visualization of rotation in the upper half-plane model of the hyperbolic plane. Link: `https://www.desmos.com/calculator/oaayzjofur`

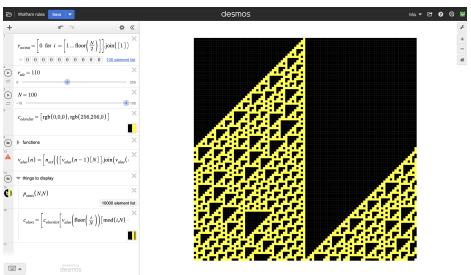

Figure 3: This graph gives a visualization of an elementary cellular automaton based on its Wolfram code. Link: `https://www.desmos.com/calculator/7d21758857`

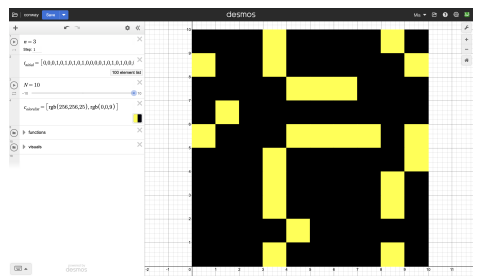

Figure 4: This graph runs Conway's Game of Life based on a starting configuration inputted by the user. Link: `https://www.desmos.com/calculator/4d1a472292`

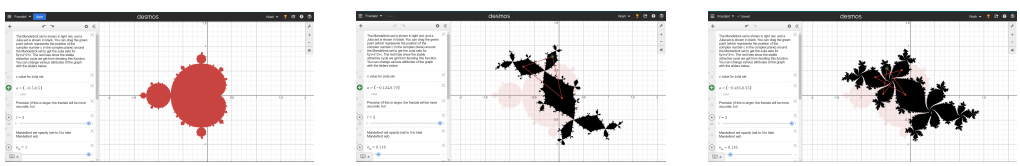

Figure 5: This interactive graph allows you to visualize several fractals. Firstly, it allows you to visualize the Mandelbrot set. Secondly, by dragging the green point around within the Mandelbrot set, you can see the filled-in Julia sets for $f(z) = z^2 + c$ for various values of the constant c (the position of the green point corresponds to the position of c in the complex plane). The red lines show the attractor cycle we get when we iterate the function $f(z) = z^2 + c$. Link: `https://www.desmos.com/calculator/ff8hdwbmws`